industrial automation

Dynamic Programming Scheduling

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Dynamic programming fundamentals

- Principle of Optimality: An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision. (See Bellman, 1957, Chap. III.3.)

Basic example

Example

$$\boxed{1 \mid\mid \sum_j \frac{T_j}{n}}$$

N. Of jobs

1 machine

Lateness

$$T_j = \max\{C_j - d_j, 0\} = \max\{L_j, 0\}$$

Tardiness

Completion time

Due date

Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

Dynamic programming fundamentals

- Required definitions:
  - State
  - Control
  - Dynamics of the system
  - Value function

State of the system for this scheduling problem

State of the system:

not ordered list (i.e. set) of complete jobs

$$X_k = \{J_{i(1)}, \ldots, J_{i(k)}\} \qquad k = 0, \ldots, n$$

$$X_0 = \emptyset$$

Example: $X_3 = \{J_1, J_2, J_4\}$

Control of the system for this scheduling problem

Control of the system at state X_k:

job executed in position (k+1)-th

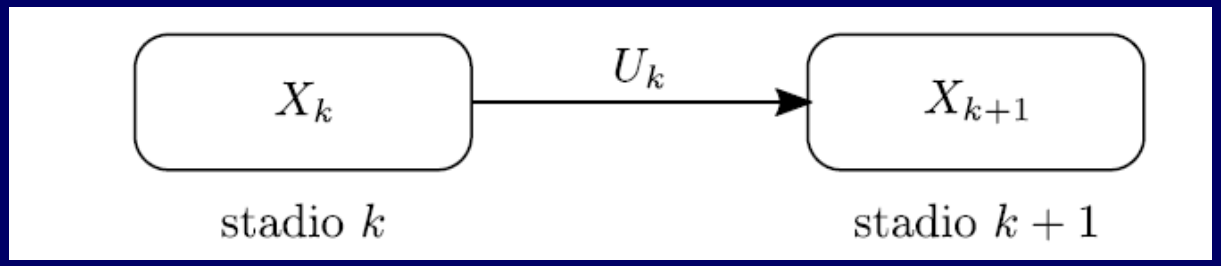
$$U_k(X_k) = J_{i(k+1)} \qquad k = 0, \ldots, n-1$$

Example: optimal control at state X_3

$$U_3^\circ\big(\{J_1, J_2, J_4\}\big) = J_3$$

Dynamics of the system

- The dynamic of the system is divided in stages, whose dynamic is given by:



- At the generic stage k, the number of possible states are:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

This is the number of combinations of n items taken k at a time

In Matlab: b = nchoosek(n,k)

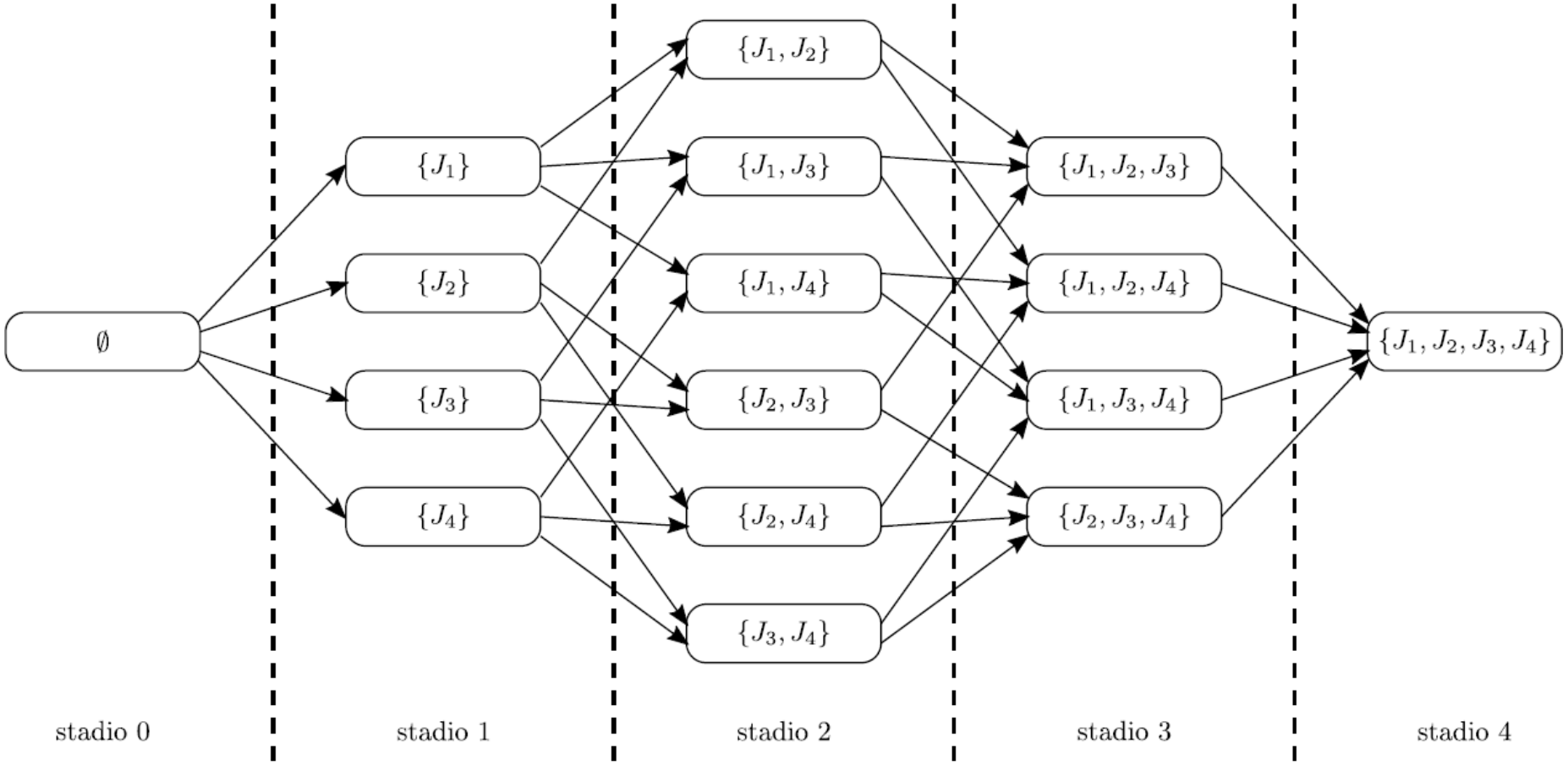
Cost of the system for this scheduling problem

Additive cost, resulting from the sum of the cost at each stage.

$$\sum_{k=0}^{n-1} H_k(X_k, U_k)$$

$$H_k(X_k, U_k) = \frac{T_{i(k+1)}}{n}$$

Application of the dynamic programming to an example



DP: Backward 1st step

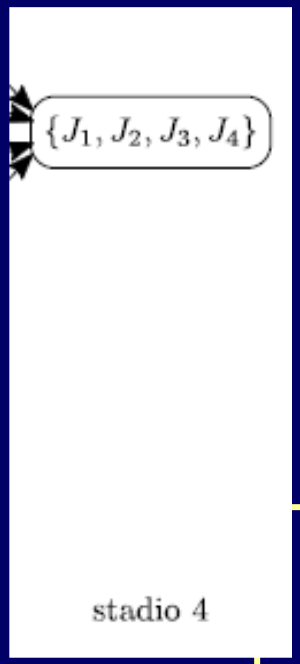
Stage 4 (the last one): no decision

Init of the cost-to-go to 0.

$$\boxed{\text{Stato } X_4 = \{J_1, J_2, J_3, J_4\}}$$

Optimal cost

$$G_4^{\circ}(\{J_1, J_2, J_3, J_4\}) = 0$$

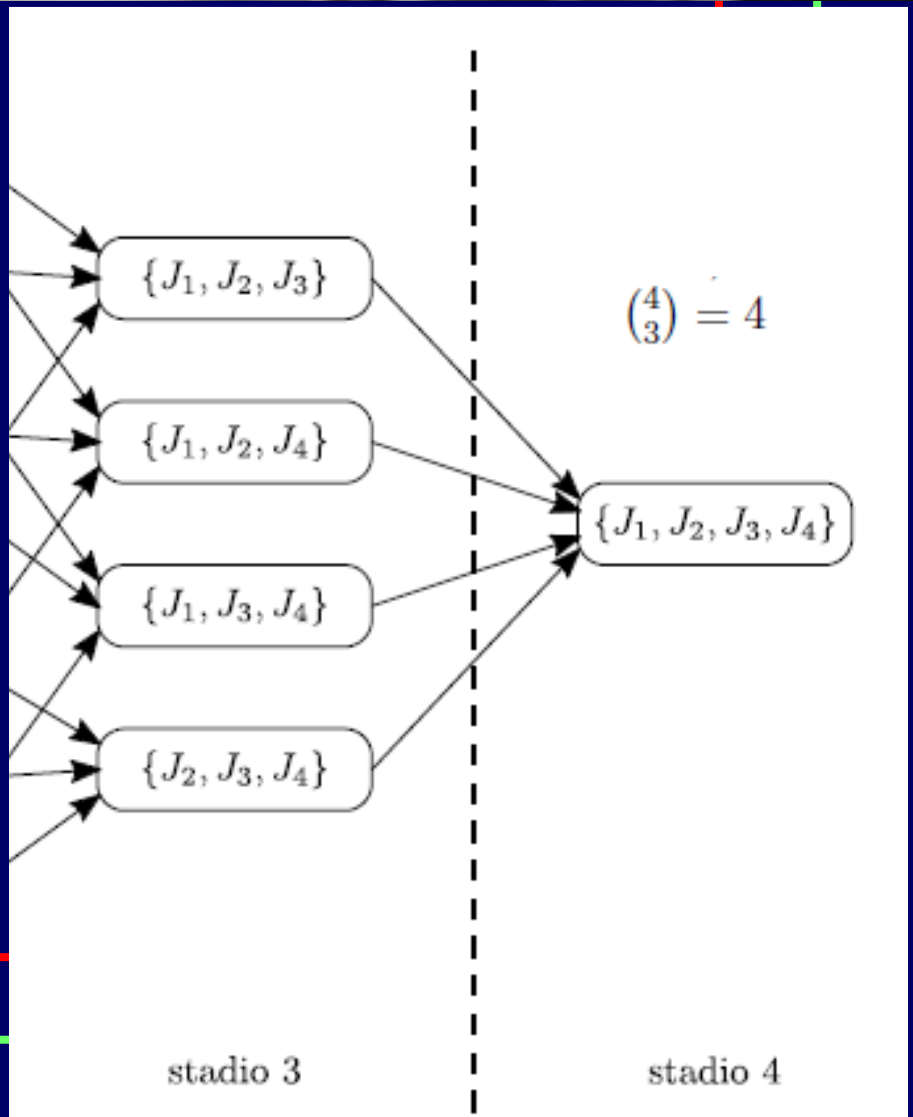


Backward 2nd step

Stage 3: decision

$\Rightarrow$ which is the last job to be executed?

Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16



Backward 2nd step

Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

$$\text{Stato } X_3 = \{J_1, J_2, J_3\}$$

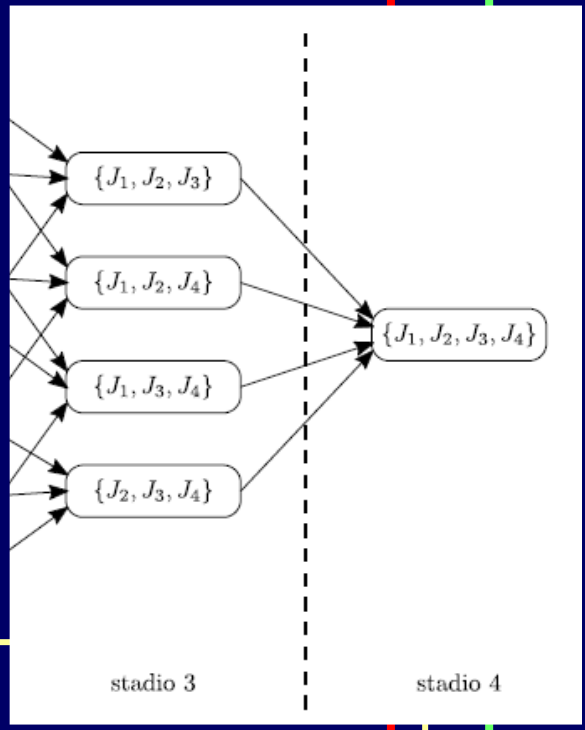
$$st_{i(4)} = ct(\{J_1, J_2, J_3\}) = p_1 + p_2 + p_3 = 8 + 6 + 10 = 24$$

$$T_4 = \max\{st_{i(4)} + p_4 - d_4, 0\} = \max\{24 + 7 - 16, 0\} = 15$$

$$G_3^\circ(\{J_1, J_2, J_3\}) = \frac{T_4}{4} + G_4^\circ(\{J_1, J_2, J_3, J_4\}) = \frac{15}{4} + 0 = \frac{15}{4}$$

$$U_3^\circ(\{J_1, J_2, J_3\}) = J_4$$

stato	cost-to-go	next job
$\{J_1, J_2, J_3\}$	$\frac{15}{4}$	J_4



Backward 2nd step

$$\text{Stato } X_3 = \{J_1, J_2, J_4\}$$

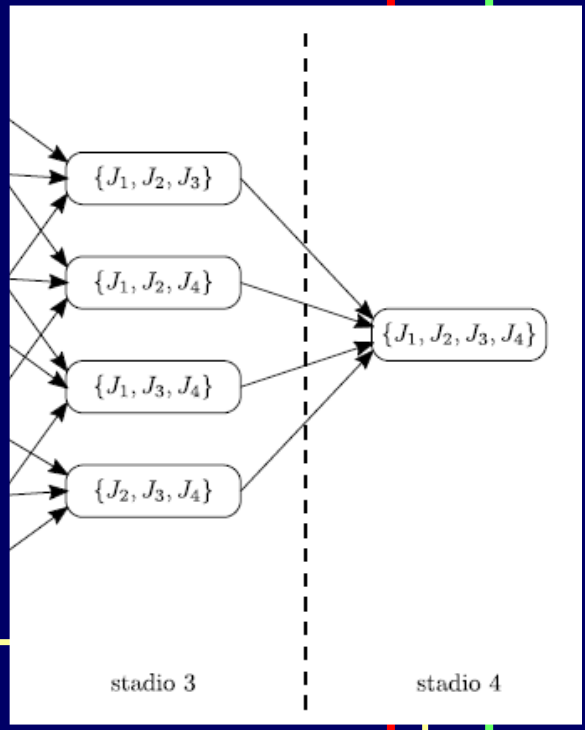
$$st_{i(4)} = ct(\{J_1, J_2, J_4\}) = p_1 + p_2 + p_4 = 8 + 6 + 7 = 21$$

$$T_3 = \max\{st_{i(4)} + p_3 - d_3, 0\} = \max\{21 + 10 - 16, 0\} = 15$$

$$G_3^\circ(\{J_1, J_2, J_4\}) = \frac{T_3}{4} + G_4^\circ(\{J_1, J_2, J_3, J_4\}) = \frac{15}{4} + 0 = \frac{15}{4}$$

$$U_3^\circ(\{J_1, J_2, J_4\}) = J_3$$

stato	cost-to-go	next job
$\{J_1, J_2, J_3\}$	$\frac{15}{4}$	J_4
$\{J_1, J_2, J_4\}$	$\frac{15}{4}$	J_3



Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

Backward 2nd step

$$\text{Stato } X_3 = \{J_1, J_3, J_4\}$$

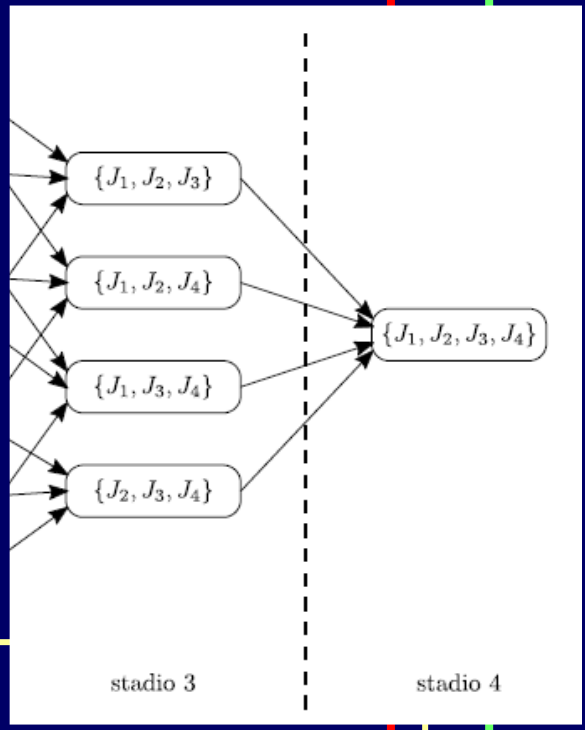
$$st_{i(4)} = ct(\{J_1, J_3, J_4\}) = p_1 + p_3 + p_4 = 8 + 10 + 7 = 25$$

$$T_2 = \max\{st_{i(4)} + p_2 - d_2, 0\} = \max\{25 + 6 - 9, 0\} = 22$$

$$G_3^\circ(\{J_1, J_3, J_4\}) = \frac{T_2}{4} + G_4^\circ(\{J_1, J_2, J_3, J_4\}) = \frac{22}{4} + 0 = \frac{11}{2}$$

$$U_3^\circ(\{J_1, J_3, J_4\}) = J_2$$

stato	cost-to-go	next job
$\{J_1, J_2, J_3\}$	$\frac{15}{4}$	J_4
$\{J_1, J_2, J_4\}$	$\frac{15}{4}$	J_3
$\{J_1, J_3, J_4\}$	$\frac{11}{2}$	J_2



Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

Backward 2nd step

Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

$$\boxed{\text{Stato } X_3 = \{J_2, J_3, J_4\}}$$

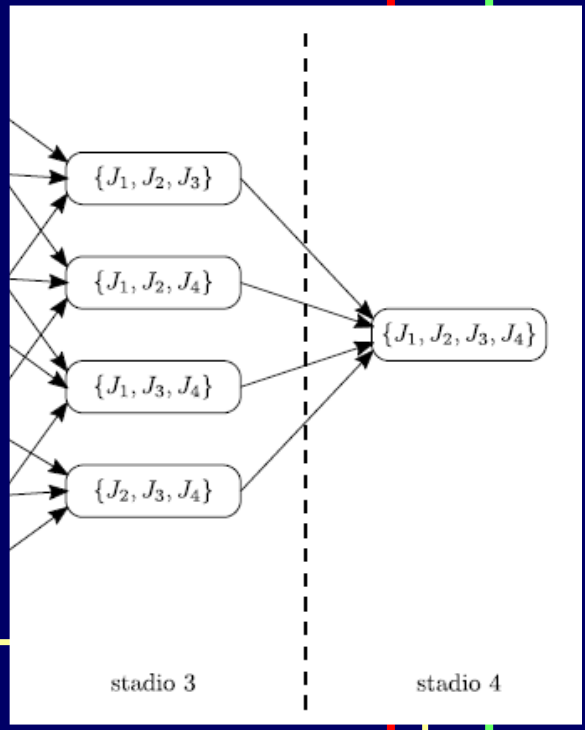
$$st_{i(4)} = ct\big(\{J_2, J_3, J_4\}\big) = p_2 + p_3 + p_4 = 6 + 10 + 7 = 23$$

$$T_1 = \max\{st_{i(4)} + p_1 - d_1, 0\} = \max\{23 + 8 - 14, 0\} = 17$$

$$G_3^\circ\big(\{J_2, J_3, J_4\}\big) = \frac{T_1}{4} + G_4^\circ\big(\{J_1, J_2, J_3, J_4\}\big) = \frac{17}{4} + 0 = \frac{17}{4}$$

$$U_3^\circ\big(\{J_2, J_3, J_4\}\big) = J_1$$

stato	cost-to-go	next job
$\{J_1, J_2, J_3\}$	$\frac{15}{4}$	J_4
$\{J_1, J_2, J_4\}$	$\frac{15}{4}$	J_3
$\{J_1, J_3, J_4\}$	$\frac{11}{2}$	J_2
$\{J_2, J_3, J_4\}$	$\frac{17}{4}$	J_1



Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

Backward 3rd step

$$\boxed{\text{Stato } X_2 = \{J_1, J_2\}}$$

$$st_{i(3)} = ct\big(\{J_1, J_2\}\big) = p_1 + p_2 = 8 + 6 = 14$$

$$T_{i(3)} = T_3 = \max\{st_{i(3)} + p_3 - d_3, 0\} = \max\{14 + 10 - 16, 0\} = 8$$

$$G_2\big(\{J_1, J_2\} \mid J_{i(3)} \equiv J_3\big) = \frac{T_3}{4} + G_3^\circ\big(\{J_1, J_2, J_3\}\big) = \frac{8}{4} + \frac{15}{4} = \frac{23}{4}$$

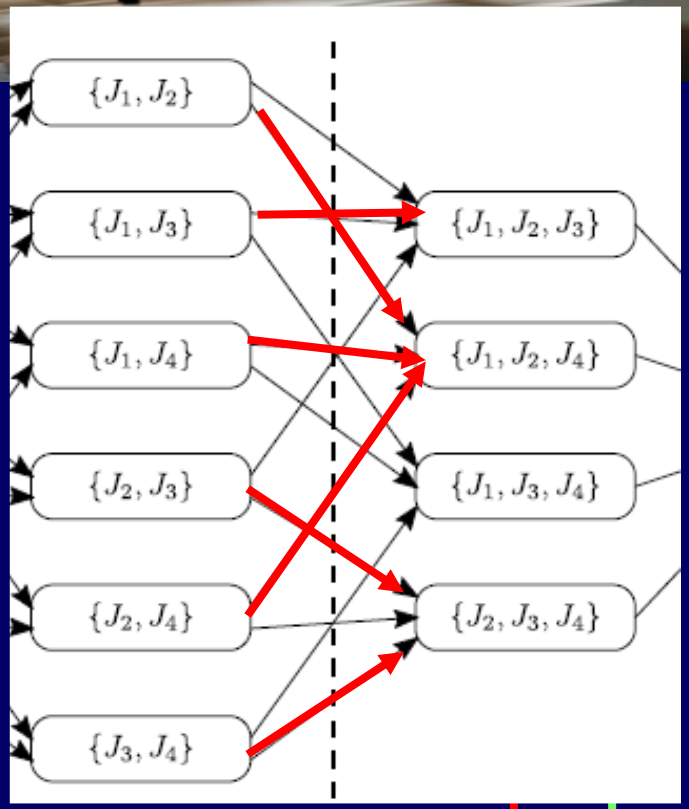
$$T_{i(3)} = T_4 = \max\{st_{i(3)} + p_4 - d_4, 0\} = \max\{14 + 7 - 16, 0\} = 5$$

$$G_2\big(\{J_1, J_2\} \mid J_{i(3)} \equiv J_4\big) = \frac{T_4}{4} + G_3^\circ\big(\{J_1, J_2, J_4\}\big) = \frac{5}{4} + \frac{15}{4} = 5$$

$$G_2^\circ\big(\{J_1, J_2\}\big) = \min\Big\{ G_2\big(\{J_1, J_2\} \mid J_{i(3)} \equiv J_3\big), G_2\big(\{J_1, J_2\} \mid J_{i(3)} \equiv J_4\big)\Big\} = \min\Big\{\frac{23}{4}, 5\Big\} = 5$$

stato	cost-to-go	next job
$\{J_1, J_2\}$	5	J_4
$\{J_1, J_3\}$	$\frac{15}{2}$	J_2
$\{J_1, J_4\}$	$\frac{27}{4}$	J_2
$\{J_2, J_3\}$	6	J_4
$\{J_2, J_4\}$	$\frac{11}{2}$	J_1
$\{J_3, J_4\}$	$\frac{31}{4}$	J_2

$$U_2^\circ\big(\{J_1, J_2\}\big) = J_4$$



Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

To be done for all states

Backward 4th step

Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

$$\boxed{\text{Stato } X_1 = \{J_1\}}$$

$$st_{i(2)} = ct(\{J_1\}) = p_1 = 8$$

$$G_1(\{J_1\} \mid J_{i(2)} \equiv J_2) = \frac{T_2}{4} + G_2^\circ(\{J_1, J_2\}) = \frac{\max\{8+6-9, 0\}}{4} + 5 = \frac{5}{4} + 5 = \frac{25}{4}$$

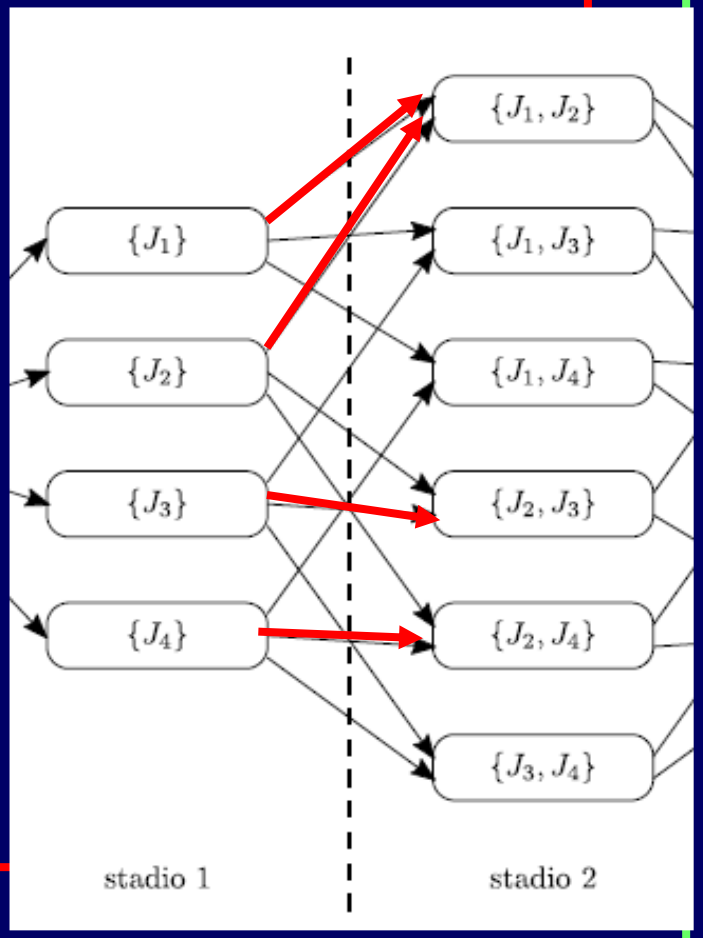
$$G_1(\{J_1\} \mid J_{i(2)} \equiv J_3) = \frac{T_3}{4} + G_2^\circ(\{J_1, J_3\}) = \frac{\max\{8+10-16, 0\}}{4} + \frac{15}{2} = \frac{2}{4} + \frac{15}{2} = 8$$

$$G_1(\{J_1\} \mid J_{i(2)} \equiv J_4) = \frac{T_4}{4} + G_2^\circ(\{J_1, J_4\}) = \frac{\max\{8+7-16, 0\}}{4} + \frac{27}{4} = 0 + \frac{27}{4} = \frac{27}{4}$$

$$G_1^\circ(\{J_1\}) = \min\left\{\frac{25}{4}, 8, \frac{27}{4}\right\} = \frac{25}{4}$$

$$U_1^\circ(\{J_1\}) = J_2$$

stato	cost-to-go	next job
$\{J_1\}$	$\frac{25}{4}$	J_2
$\{J_2\}$	5	J_1
$\{J_3\}$	$\frac{31}{4}$	J_2
$\{J_4\}$	$\frac{13}{2}$	J_2



To be done for all states

Job	J_1	J_2	J_3	J_4
p_i	8	6	10	7
d_i	14	9	16	16

Backward 5th step

$$\boxed{\text{Stato } X_0 = \emptyset}$$

$$st_{i(1)} = 0$$

$$G_0(\emptyset \mid J_{i(1)} \equiv J_1) = \frac{T_1}{4} + G_1^\circ(\{J_1\}) = \frac{\max\{0+8-14,0\}}{4} + \frac{25}{4} = 0 + \frac{25}{4} = \frac{25}{4}$$

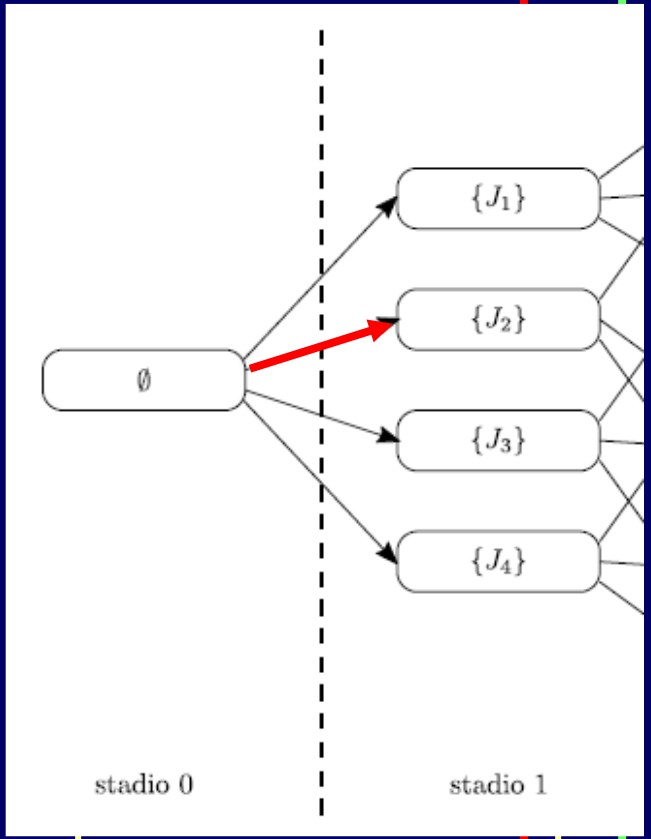
$$G_0(\emptyset \mid J_{i(1)} \equiv J_2) = \frac{T_2}{4} + G_1^\circ(\{J_2\}) = \frac{\max\{0+6-9,0\}}{4} + 5 = 0 + 5 = 5$$

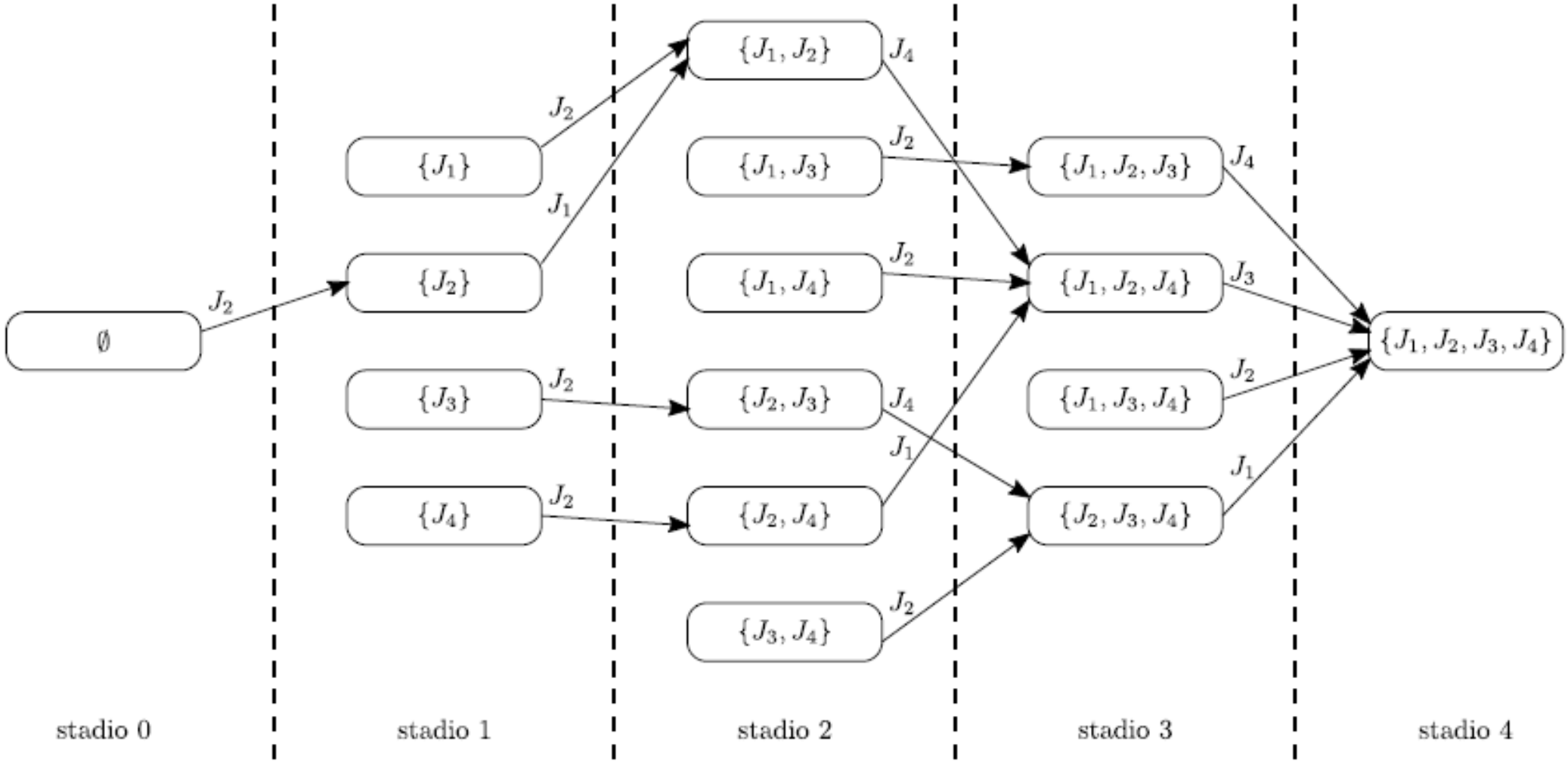
$$G_0(\emptyset \mid J_{i(1)} \equiv J_3) = \frac{T_3}{4} + G_1^\circ(\{J_3\}) = \frac{\max\{0+10-16,0\}}{4} + \frac{17}{4} = 0 + \frac{31}{4} = \frac{31}{4}$$

$$G_0(\emptyset \mid J_{i(1)} \equiv J_4) = \frac{T_4}{4} + G_1^\circ(\{J_4\}) = \frac{\max\{0+7-16,0\}}{4} + \frac{13}{2} = 0 + \frac{13}{2} = \frac{13}{2}$$

$$G_0^\circ(\emptyset) = \min\left\{\frac{25}{4}, 5, \frac{31}{4}, \frac{13}{2}\right\} = 5$$

$$U_0^\circ(\emptyset) = J_2$$

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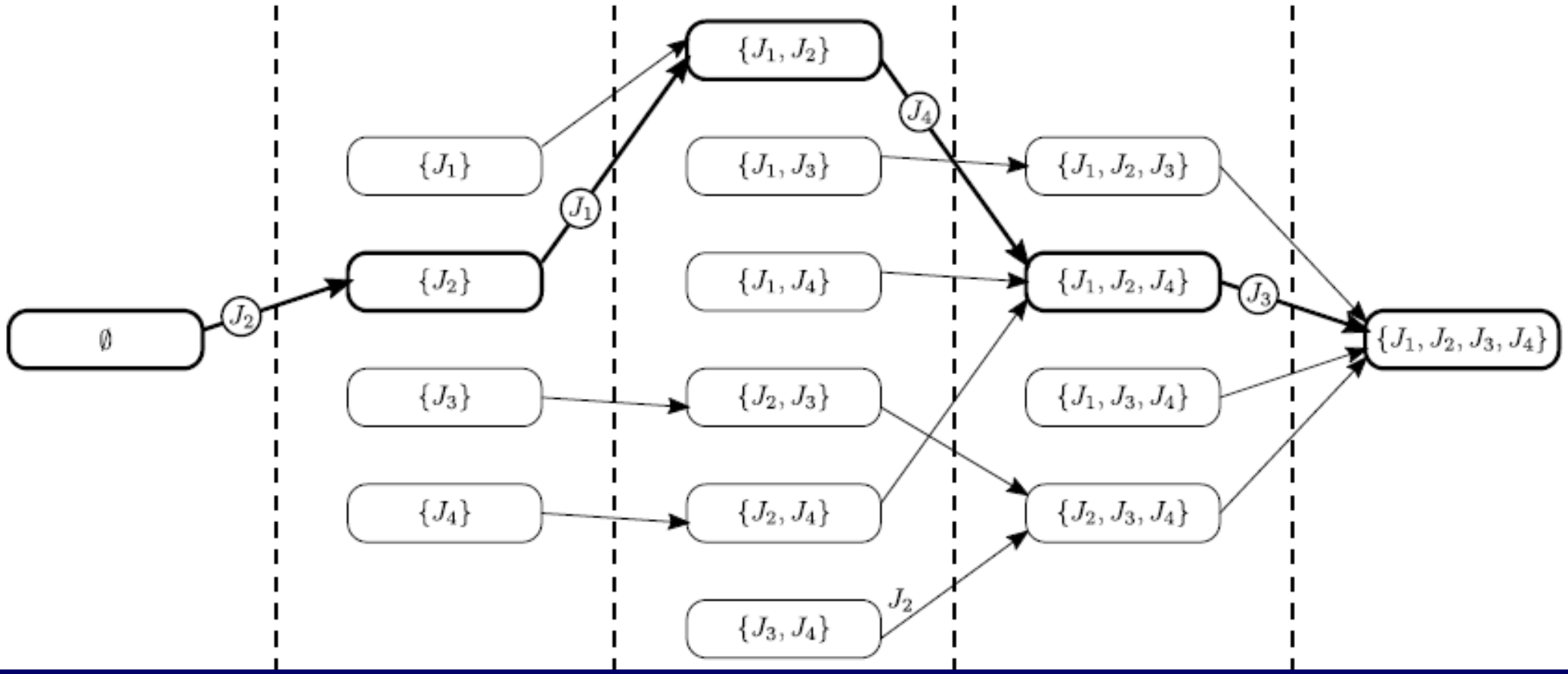
Forward phase

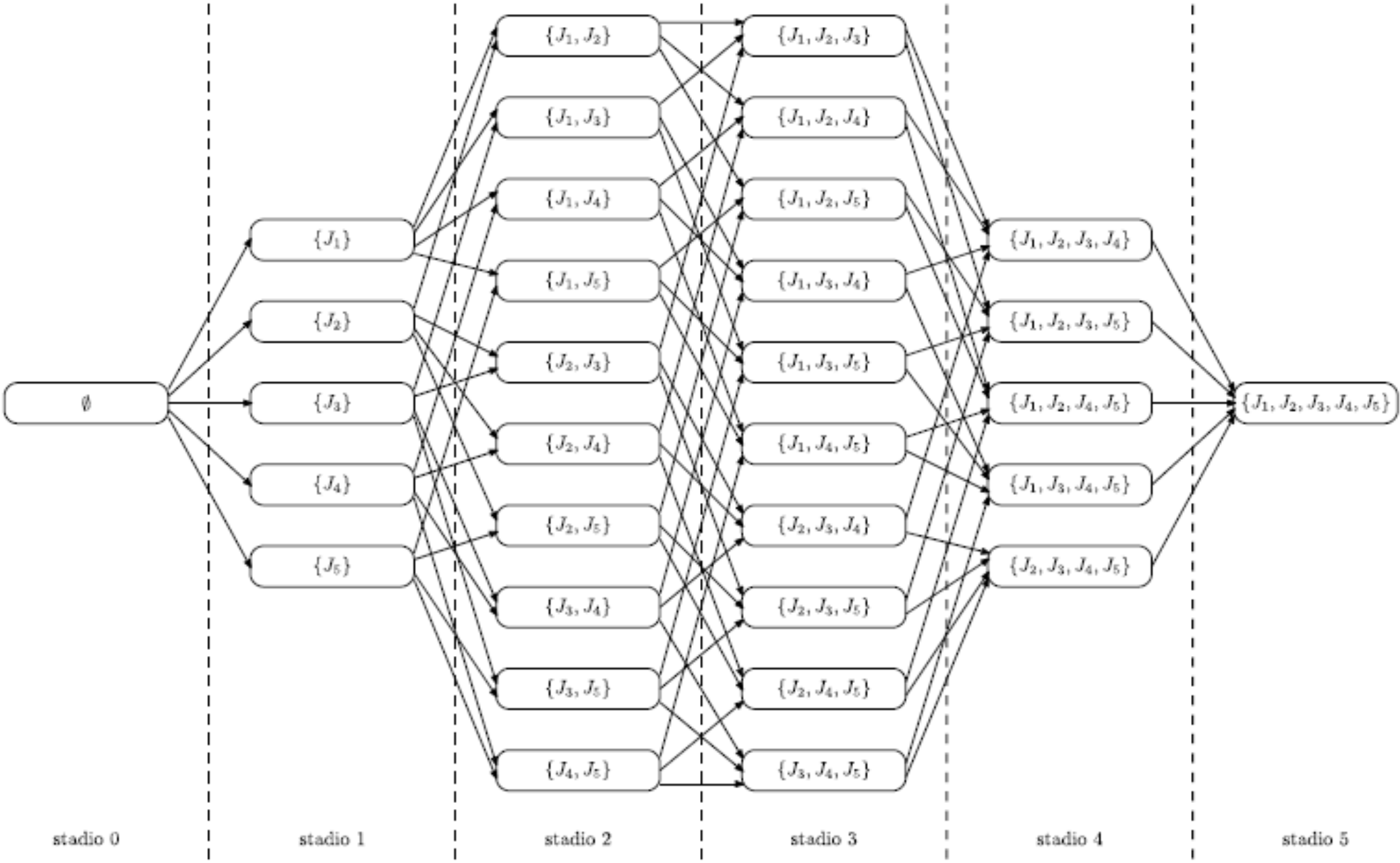
Optimal solution $J_2 \rightarrow J_1 \rightarrow J_4 \rightarrow J_3$ **Average tardness 5**



New scheduling problem

$$1 \mid \textbf{prec} \mid \sum_j w_j T_j$$

Job	J_1	J_2	J_3	J_4	J_5
p_i	4	2	6	3	5
d_i	7	9	9	6	12
w_i	1.5	1.5	1.5	1	1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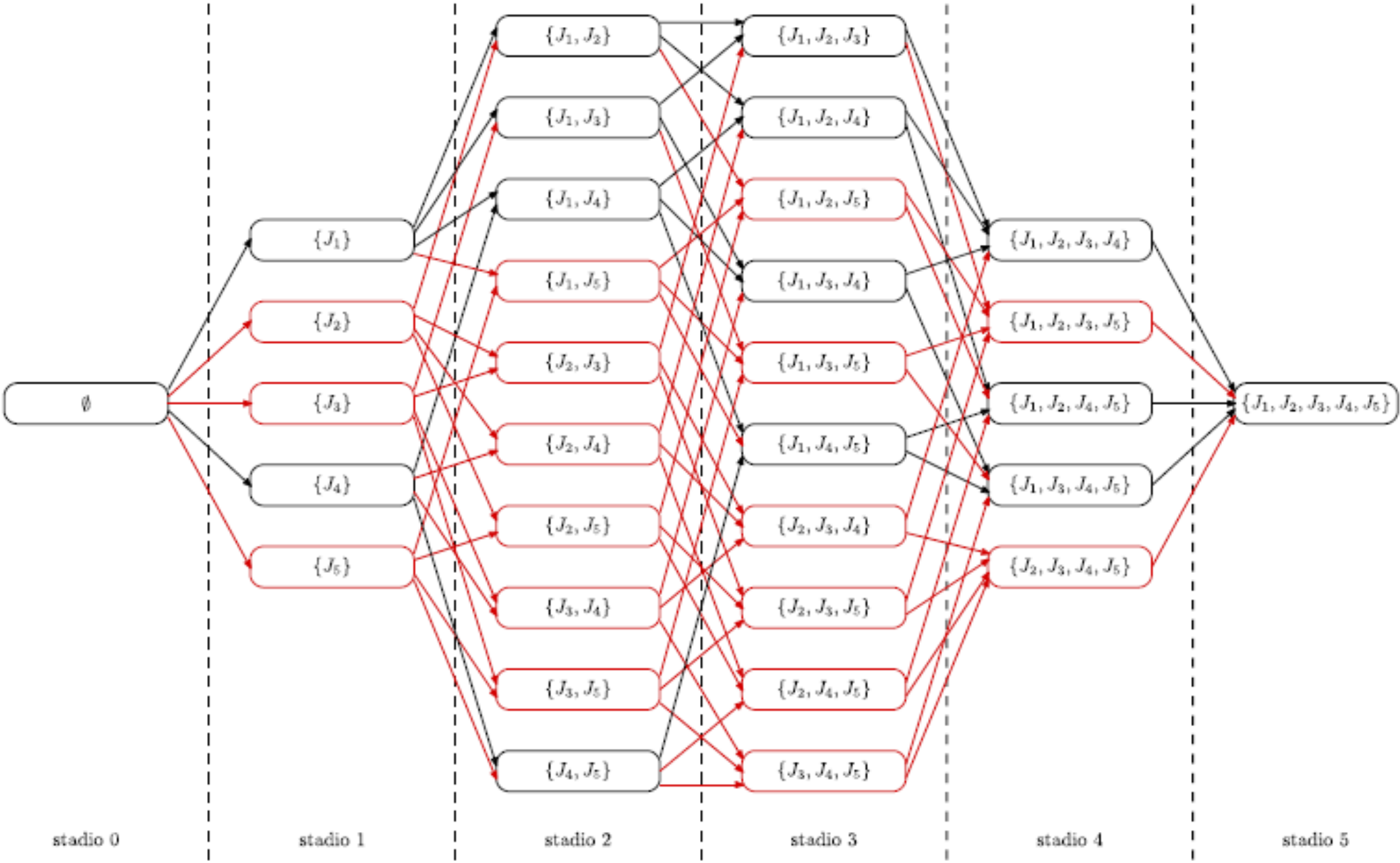
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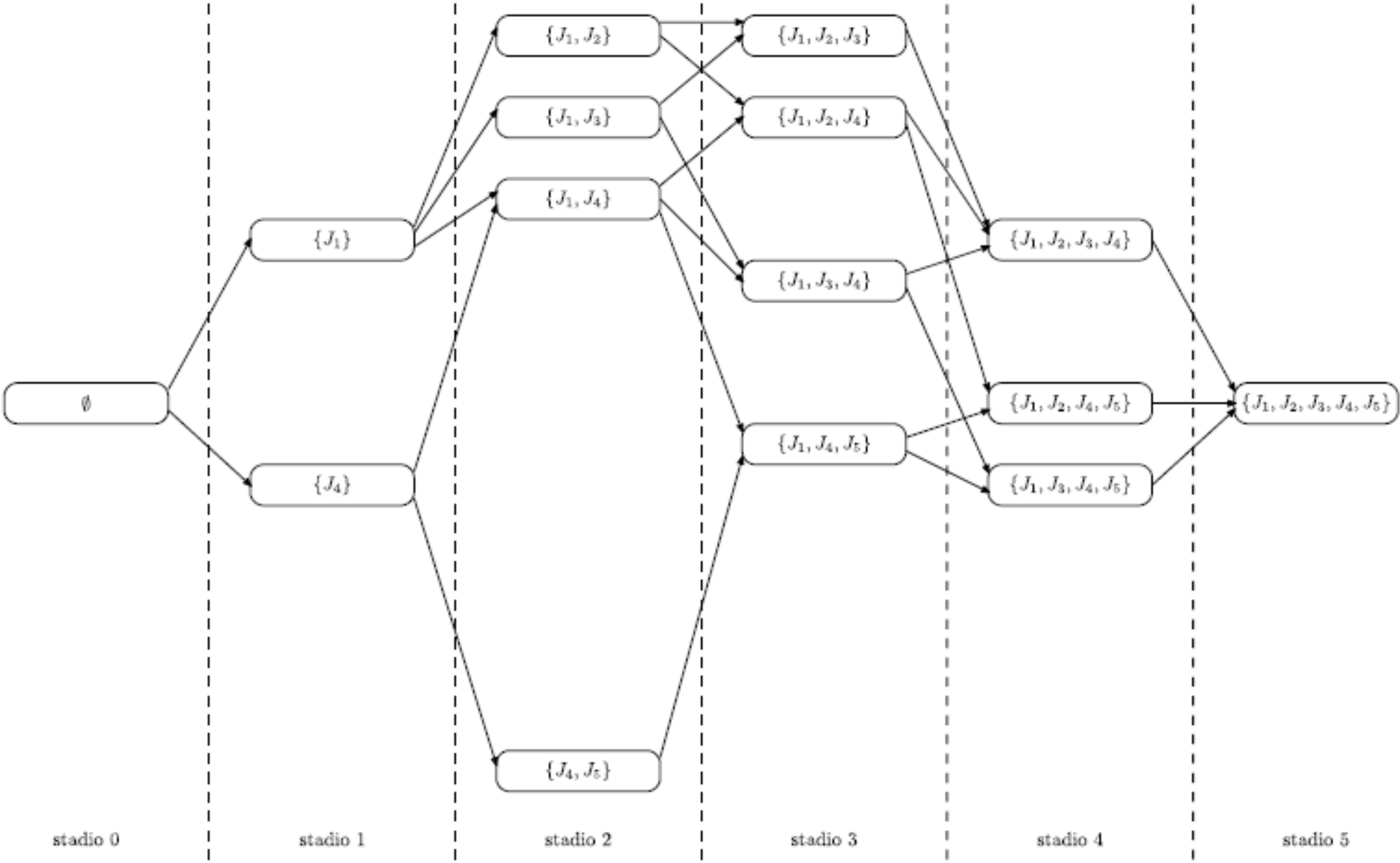
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